# Supervisor's Final Words

BDS–SMC2 Node — Pre-Submission Assessment

Examiner / Supervisor review of
*BDS-SMC2 Node — Comprehensive Report* (WP5, BDS Communication)
Candidate: Letsoalo Maile

June 29, 2026

**Verdict**

This is a **strong submission**. The candidate has done what most fail to do: separated the *assigned deliverable* (WP5 — complete and verified in the group ROS 2 system) from the *research beside it* (hardware bring-up, field campaign, payload redesign), and bounded the research honestly. I would pass this comfortably and short-list it for distinction, *provided the four propagation fixes below are closed.*

## What earns the marks

- A concrete, git-backed before/after capability narrative — not hand-waving.
- Every figure and statistic traced to a committed script and a CSV; the reproduce-everything appendix is exactly what a panel wants.
- The candidate names and bounds their own weaknesses (A1, A2, B1, B2) before the panel can. A student who has already bounded every limitation cannot be ambushed.
- The consolidated hardware record (§7.4, Figure H) places the unrequested, simulation-first-exceeding hardware work in one verifiable picture.

## The four fixes required before submission

All cheap, all free marks.

1. **Propagate the diagnosed fixes (C1–C4).** The plan was ahead of the artifacts; the prose, README, and figures must now actually say what the audit already concluded. *(C1, C2, C4 closed in text; C3 closed in prose — Figure 3's hard 210-bit line still needs regenerating from its plotting script.)*
2. **State verb discipline in the main body.** "We demonstrate" requires hardware; "we show in simulation / establish the bound" is unattackable.
3. **The single hardware day is the highest-leverage hour.** One real 112-bit transmission plus a portal screenshot upgrades every "simulate" to "demonstrate."
4. **Section numbering tidied** (§10.x, §12.x corrected) — meticulous work should not look careless.

## On the UN Sustainable Development Goals

Lead with the goals the results *demonstrate*; present the rest as alignment.

| SDG | Claim | Basis |
|---|---|---|
| **11** Sustainable Cities & Communities (11.5, 11.b) | Primary | Disaster-resilient survivor channel; reliability & latency results. |
| **9** Industry, Innovation & Infrastructure (9.1, 9.c) | Primary | BeiDou-3 SMC as resilient ICT + novel payload + autonomous pipeline. |
| **3** Good Health & Well-being (3.6, 3.d) | Supporting | Shorter rescue loop; outcome-level, not directly measured. |
| **13** Climate Action (13.1) | Contextual | Climate-driven hazards; framing, not a direct result. |

**Discipline note:** claim SDG 11 and SDG 9 as demonstrated; claim SDG 3 and SDG 13 as alignment, not proven impact. "Lives saved" would be the one overclaim in an otherwise disciplined document.

## Responsibilities vs achievement (the closing verdict)

| Proposal responsibility (WP5) | Outcome |
|---|---|
| BDS-SMC emulator node | **Met** |
| Coordinate-message parsing | **Exceeded** (3 formats) |
| Auto-inject → precise PNT | **Met** |
| QGC / mission interface | **Met** |
| ROS 2 integration (Phase 3) | **Met & verified** |

Every assigned responsibility was met; two were exceeded. Beyond the simulation-first brief the candidate delivered real hardware, a 232-transmission reliability campaign, a latency baseline, and the 64→112-bit payload research. One honest boundary remains: the 112-bit frame is software- and simulation-validated but not yet satellite-flown.

**Bottom line**

The work is done. The distance from *strong chapter* to *publishable* is one hardware transmission and an afternoon of propagating fixes already identified. Close those and submit with confidence.

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*Supervisor's signature:* ____________________ Date: ____________